mathongo

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir

JEE Mains 2020 Jan Chapter wise Question Bank

Differentiation

Q1

Let माना $(x)^k + (y)^k = (a)^k$ where जहां a, $k > 0$ and तथा $\frac{dy}{dx} + \left(\frac{y}{x}\right)^{\frac{1}{3}} = 0$, then find तब k बराबर है–

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$ (3) $\frac{4}{3}$ (1) 2

7th Jan Morning

Sol

(2)

$$k.x^{k-1} + k.y^{k-1}\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\left(\frac{x}{y}\right)^{k-1}$$

$$\frac{dy}{dx} + \left(\frac{x}{y}\right)^{k-1} = 0$$

$$k - 1 = -\frac{1}{3}$$

$$k = 1 - \frac{1}{3} = \frac{2}{3}$$

02

If यदि $y\sqrt{1-x^2} = k - x\sqrt{1-y^2}$ and और $y\left(\frac{1}{2}\right) = -\frac{1}{4}$. Then तो $\frac{dy}{dx}$ at $x = \frac{1}{2}$ is होगा -

(1) $\frac{2}{\sqrt{5}}$ (2) $-\frac{\sqrt{5}}{4}$ (3) $-\frac{\sqrt{5}}{2}$ (4) $\frac{\sqrt{5}}{2}$

7th Jan Evening

Sol

(3)

$x = \frac{1}{2},\ y = \frac{-1}{4} \Rightarrow \quad xy = \frac{-1}{8}$

$$y.\frac{1.(=2x)}{2\sqrt{-x^2}} + y'.\sqrt{1-x^2} = -\left\{1.\sqrt{1-y^2} + \frac{x.(-2y)}{2\sqrt{1-y^2}}y'\right\}$$

$$-\frac{xy}{\sqrt{1-x^2}} + y'\sqrt{1-x^2} = -\sqrt{1-y^2} + \frac{xy.y'}{\sqrt{1-y^2}}$$

$$y'\left(\sqrt{1-x^2} - \frac{xy}{\sqrt{1-y^2}}\right) = \frac{xy}{\sqrt{1-x^2}} - \sqrt{1-y^2}$$

$$y'\left(\frac{\sqrt{3}}{2} + \frac{1}{8.\frac{\sqrt{15}}{4}}\right) = \frac{-1}{8.\sqrt{\frac{3}{2}}} - \frac{\sqrt{15}}{4}$$

$$y'\left(\frac{\sqrt{45}+1}{2\sqrt{15}}\right) = -\frac{\left(1+\sqrt{45}\right)}{4\sqrt{3}}$$

$$y' = -\frac{\sqrt{5}}{2}$$

03

Let $f(x) = \{(\sin(\tan^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1$ where $|x| > 1$ and $\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}\left(\sin^{-1} f(x)\right)$.

If $y\left(\sqrt{3}\right) = \frac{\pi}{6}$ then $y\left(-\sqrt{3}\right) =$

माना $f(x) = \{(\sin(\tan^{-1}x) + \sin(\cot^{-1}x)\}^2 - 1$ जहाँ $|x| > 1$ तथा $\frac{dy}{dx} = \frac{1}{2}\frac{d}{dx}\left(\sin^{-1} f(x)\right)$.

यदि $y\left(\sqrt{3}\right) = \frac{\pi}{6}$ तब $y\left(-\sqrt{3}\right) =$

(1) $\frac{5\pi}{6}$ (2) $\frac{-\pi}{6}$ (3) $\frac{\pi}{3}$ (4) $\frac{2\pi}{3}$

8th Jan Morning

Sol

(2)

$2y = \sin^{-1}f(x) + C = \sin^{-1}(\sin(2\tan^{-1}x)) + C \quad \Rightarrow \quad 2\left(\frac{\pi}{6}\right) = \sin^{-1}\left(\sin\left(\frac{2\pi}{3}\right)\right) + C$

$\frac{\pi}{3} = \frac{\pi}{3} + C \quad \therefore \quad C = 0$

for $x = -\sqrt{3}$, $\quad 2y = \sin^{-1}\left(\sin\left(\frac{-2\pi}{6}\right)\right) + 0 \quad \Rightarrow \quad 2y = \frac{-\pi}{3}$

$\left(y = \frac{-\pi}{6}\right)$

04

Let $x = 2\sin\theta - \sin2\theta$ and
$y = 2\cos\theta - \cos2\theta$

find $\frac{d^2y}{dx^2}$ at $\theta = \pi$

माना $x = 2\sin\theta - \sin2\theta$ तथा
$y = 2\cos\theta - \cos2\theta$
$\theta = \pi$ पर $\frac{d^2y}{dx^2}$ का मान ज्ञात कीजिये।

(1) $\frac{3}{8}$ (2) $\frac{3}{2}$ (3) $\frac{5}{8}$ (4) $\frac{7}{8}$

9th Jan Evening

Sol

$\frac{dx}{d\theta} = 2\cos\theta - 2\cos2\theta$

$\frac{dy}{d\theta} = -2\sin\theta + 2\sin2\theta$

$\therefore \frac{dy}{dx} = \frac{\sin2\theta - \sin\theta}{\cos\theta - \cos2\theta}$

$= \frac{2\sin\frac{\theta}{2}.\cos\frac{3\theta}{2}}{2\sin\frac{\theta}{2}\sin\frac{3\theta}{2}} = \cot\frac{3\theta}{2}$

$$\frac{d^2y}{dx^2} = \frac{-3}{2}\text{cosec}^2\frac{3\theta}{2}.\frac{d\theta}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{-\frac{3}{2}\text{cosec}^2\frac{3\theta}{2}}{2(\cos\theta - \cos 2\theta)}$$

$$\frac{d^2y}{dx^2}_{(\theta=\pi)} = -\frac{3}{4(-1-1)} = \frac{3}{8}$$